

(*) All the rings will be assumed to be a commutative ring with identity if there is no special statement. In this section, the module M will be assumed to be an A -module.

Example 0.1. : If A is a field, then the A -module M is precisely a vector space over A .

Example 0.2. : For a ring A , if M is an ideal of A , then M is an A -module with the scalar multiplication

$$\begin{aligned} A \times M &\rightarrow M \\ (a, m) &\mapsto a \cdot m := am \end{aligned}$$

Example 0.3. : For a ring A , $A[x]$ is an A -module with the usual scalar multiplication on polynomials.

Example 0.4. : Suppose I is an ideal of A , then A/I is an A -module with the scalar multiplication

$$\begin{aligned} A \times A/I &\rightarrow A/I \\ (a, x + I) &\mapsto a \cdot (x + I) := ax + I \end{aligned}$$

Example 0.5. : Let V be a vector space over a field F . $T : V \rightarrow V$ is a linear operator on V . For $\forall p(x) \in F[x]$, define

$$\begin{aligned} F[x] \times V &\rightarrow V \\ (p(x), v) &\mapsto p(x)v := p(T)(v) \end{aligned}$$

then it is easy to check that V is an $F[x]$ -module with respect to this scalar multiplication.

Example 0.6. : Let $(G, +)$ be an abelian group, then G is a $\mathbb{Z}$ -module with

$$\begin{aligned} \mathbb{Z} \times G &\rightarrow G \\ (n, g) &\mapsto ng := g + g + \cdots + g \text{ (n times)}. \end{aligned}$$

recall the definition of submodule

Definition 0.1. : A subset $N \subseteq M$ is called an **A -submodule of M** if N is also an A -module w.r.t(i.e., with respect to) the same addition and scalar multiplication of M .

recall the definition of module homomorphism

Definition 0.2. : Let M and N be two A -modules, a map $f : M \rightarrow N$ is called an **A -module homomorphism** if

- (1) $f(m_1 + m_2) = f(m_1) + f(m_2)$ ($\forall m_1, m_2 \in M$)
- (2) $f(am) = af(m)$ ($\forall a \in A, \forall m \in M$)

Example 0.7. :

$$f : \mathbb{Z} \rightarrow \mathbb{Z}$$

$$n \mapsto mn$$

is a $\mathbb{Z}$ -module homomorphism with m a fixed integer.

Example 0.8. : Let $A = \mathbb{R}$, $M = C^1[a, b]$ and $N = C[a, b]$, define

$$\varphi : M \rightarrow N$$

$$f(x) \mapsto \frac{d}{dx} f(x)$$

then φ is an A -module homomorphism.

Example 0.9. : Let M, N be two A -modules, define $\text{Hom}_A(M, N) := \{f : M \rightarrow N \mid f \text{ is a homomorphism}\}$ and define the scalar multiplication as

$$A \times \text{Hom}_A(M, N) \rightarrow \text{Hom}_A(M, N)$$

$$(a, f) \mapsto af,$$

where $(af)(m) := a(f(m))$ ($\forall m \in M$) and the addition on $\text{Hom}_A(M, N)$ is the usual addition of homomorphism. Then $\text{Hom}_A(M, N)$ is an A -module w.r.t the defined addition and scalar multiplication.

Example 0.10. :

$$\varphi : \text{Hom}_{\mathbb{Z}}(\mathbb{Z}, \mathbb{Z}) \rightarrow \mathbb{Z}$$

$$f \mapsto f(1)$$

is an isomorphism.

Example 0.11. :

$$\varphi : \text{Hom}_A(A, M) \rightarrow M$$

$$f \mapsto f(1)$$

is an isomorphism.

Example 0.12. : Let $f : M \rightarrow M'$ be an A -module homomorphism and N is another A -module, define

$$\varphi : \text{Hom}_A(M', N) \rightarrow \text{Hom}_A(M, N)$$

$$h \mapsto \varphi(h) := h \circ f$$

i.e., we have the commutative diagram

$$\begin{array}{ccc} M & \xrightarrow{f} & M' \\ & \searrow h \circ f & \downarrow h \\ & & N \end{array}$$

then it's easy to check that φ is an A -module homomorphism.

Example 0.13. :If we define

$$\varphi : \text{Hom}_A(N, M) \rightarrow \text{Hom}_A(N, M')$$

$$h \mapsto f \circ h$$

i.e., we have the diagram

$$\begin{array}{ccc} M & \xrightarrow{f} & M' \\ & \nwarrow h & \uparrow f \circ h \\ & & N \end{array}$$

then φ is also an A -module homomorphism.

Suppose N is an A -submodule of M , define a relation $\sim$ on M :

$$x \sim y \Leftrightarrow x - y \in N,$$

then we can check that the relation $\sim$ is actually an equivalence relation on M . The set of all equivalence class is denoted by M/N , i.e, $M/N = \{x + N | x \in M\}$, then M/N is an A -module with

$$(x + N) + (y + N) := (x + y) + N$$

$$a(x + N) := ax + N \quad (\forall a \in A)$$

Isomorphism Theorem

Theorem 0.1. :Let $f : M \rightarrow N$ be an A -module homomorphism, $\ker f := \{m \in M | f(m) = 0\}$ and $\text{Im} f := \{f(m) | m \in M\}$, then we have

$$\text{Im} f \cong M / \ker f.$$

Proof: Define

$$\varphi : M / \ker f \rightarrow \text{Im} f$$

$$x + \ker f \mapsto f(x)$$

then we can check that φ is well-defined and is an isomorphism. $\square$

Remark 0.1. : (1) It is clear that $\ker f$ is a submodule of M and $\text{Im} f$ is a submodule of N ;
 (2) define $\text{coker} f := N / \text{Im} f$, then we have

$$f \text{ is injective} \Leftrightarrow \ker f = \{0\}, \quad f \text{ is surjective} \Leftrightarrow \text{coker} f = \{0\}.$$

Proposition 0.1. : Suppose $M_1 \subseteq M_2 \subseteq M$ are A -modules, i.e., M_1 is a submodule of M_2 and M_2 is a submodule of M , then we have

$$(M/M_1)/(M_2/M_1) \cong M/M_2.$$

Proof: Define

$$\varphi : M/M_1 \rightarrow M/M_2$$

$$x + M_1 \mapsto x + M_2$$

then we can check that φ is well-defined, $\ker \varphi = M_2/M_1$ and φ is surjective. Hence by the theorem above we have $(M/M_1)/(M_2/M_1) \cong M/M_2$. $\square$

recall the definition of sum of submodules

Definition 0.3. :If $\{M_i\}_{i \in I}$ is a family of A -submodules of M , then $\sum_{i \in I} M_i := \{\sum_{i \in I} m_i (\text{finite sum}) | m_i \in M_i, \forall i \in I\}$ is again a submodule of M .

Proposition 0.2. :If M_1 and M_2 are two submodules of M , then

$$(M_1 + M_2)/M_1 \cong M_2/(M_1 \cap M_2).$$

Proof: Define

$$\varphi : M_2 \rightarrow (M_1 + M_2)/M_1$$

$$x \mapsto x + M_1,$$

then $\ker \varphi = M_1 \cap M_2$ and φ is surjective. Apply the isomorphism theorem. $\square$

Definition 0.4. :Suppose M is an A -module and $S \subseteq A$ is a subset, then define

$$SM := \{\sum_i a_i x_i (\text{finite sum}) | a_i \in S, x_i \in M\}$$

Let I be the ideal generated by the subset $S \subseteq A$, i.e., $I = \langle S \rangle = \{\sum_j a_j s_j (\text{finite sum}) | a_j \in A, s_j \in S\}$, then it's clear that $IM \supseteq SM$; for $\forall \sum_i a_i x_i \in IM$, where $a_i \in I$ and $x_i \in M$, since $I = \langle S \rangle$, then a_i is of the form $a_i \triangleq \sum_j \alpha_{ij} s_{ij} (\forall i)$, where $s_{ij} \in S$ and $\alpha_{ij} \in A$. Hence $\sum_i a_i x_i = \sum_i (\sum_j \alpha_{ij} s_{ij}) x_i = \sum_{i,j} s_{ij} (\alpha_{ij} x_i) \in SM$ (note that $\alpha_{ij} x_i \in M$), thus we get $IM = \langle S \rangle M \subseteq SM$ and $\langle S \rangle M = SM$.

Hence we have the proposition

Proposition 0.3. :For an A -module M and subset $S \subseteq A$, we have $SM = \langle S \rangle M$.

Remark 0.2. : In the case that S is a set of single element $S = \{a\}$, then $SM = aM = \{am | m \in M\} (= \langle a \rangle M)$. It is a straightforward verification that $SM = \langle S \rangle M$ is an A -submodule of M .

Colon of Modules

Definition 0.5. :Let N and P be two A -submodules of M , define

$$N : P := \{a \in A \mid aP \subseteq N\} = \{a \in A \mid ap \in N(\forall p \in P)\}.$$

By the definition above, we can know that $0 \in N : P$. If $x, y \in N : P$, then $x + y \in N : P$ and for $\forall a \in A$, $(ax)p = a(xp) \in N(\forall p \in P)$, hence $ax \in N : P$, which means that $N : P$ is an ideal of A .

Particularly, it gives the definition

Annihilator of Modules

Definition 0.6. :If M is an A -module, then

$$(0) : M = \{a \in A \mid am = 0(\forall m \in M)\}$$

is called the **annihilator of M** , denoted by $Ann(M)$.

Consider the A -module M , suppose I is an ideal of A . If $I \subseteq Ann(M)$, then M is an A/I -submodule with the scalar multiplication

$$A/I \times M \rightarrow M$$

$$(\bar{x}, m) \mapsto \bar{x}m := xm$$

where $\bar{x}$ stands for $x + I$.

Proof: If $\bar{x} = \bar{y} \in A/I$, which means $x - y \in I \subseteq Ann(M)$, thus $(x - y)m = 0$ ($\forall m \in M$), i.e., $xm = ym$ and hence $\bar{x}m = \bar{y}m$. The scalar multiplication is well-defined. $\square$

Generating Set of Modules

Definition 0.7. :A set of elements $S = \{x_\lambda \mid \lambda \in \Lambda\} \subseteq M$ is said to be a **generating set of M** if

$$M = \left\{ \sum_{\lambda \in \Lambda} a_\lambda x_\lambda (\text{finite sum}) \mid a_\lambda \in A(\forall \lambda \in \Lambda) \right\},$$

particularly, if S is a finite set, then M is said to be **finitely generated**.

Example 0.14. :Let $A = k[x]$, where k is a field, $I = \{p(x) \mid p(0) = 0\}$, then $I = \langle x \rangle$.

Example 0.15. :Let $A = k$ and k is a field, then $M = k[x]$ is not finitely generated.

Suppose $\{M_\lambda\}_{\lambda \in \Lambda}$ is a collection of A -modules, define

$$\bigoplus_{\lambda \in \Lambda} M_\lambda := \{(x_\lambda)_{\lambda \in \Lambda} \mid x_\lambda = 0 \text{ for all but finitely many } \lambda\}$$

$$\prod_{\lambda \in \Lambda} M_\lambda := \{(x_\lambda)_{\lambda \in \Lambda} \mid \lambda \in \Lambda\}$$

it's clear that $\bigoplus_{\lambda \in \Lambda} M_\lambda = \prod_{\lambda \in \Lambda} M_\lambda$ if Λ is a finite index set.

Free Modules

Definition 0.8. :An A -module M is said to be **free** if $M \cong \bigoplus_{\lambda \in \Lambda} M_\lambda$ with $M_\lambda \cong A$ for all $\lambda \in \Lambda$. In other words, $M \cong A^{(\oplus \Lambda)}$.

Example 0.16. : $M = A \oplus A \triangleq A^{(2)}$ is a free A -module (with componentwise addition and scalar multiplication).

Example 0.17. :For a fixed integer $m \in \mathbb{Z}$,

$$\varphi : \mathbb{Z} \rightarrow m\mathbb{Z}$$

$$n \mapsto mn$$

is an isomorphism, which means that $m\mathbb{Z}$ is a free $\mathbb{Z}$ -module.

Example 0.18. :Let the ring A be a PID, if M is an ideal of A , then the A -module $M \cong A$, i.e., M is a free A -module.

Example 0.19. :Let $A = \mathbb{Z}$, $M = \mathbb{Z}[x]$, then $M \cong \mathbb{Z}^{(\oplus \mathbb{N})}$, i.e., $\mathbb{Z}[x]$ is a free $\mathbb{Z}$ -module.

Example 0.20. : $\mathbb{Z}_2$ is not a free $\mathbb{Z}$ -module (Since $\bigoplus^\Lambda \mathbb{Z} \rightarrow \mathbb{Z}_2$ is not injective for any index set Λ). Generally, any finite abelian group is not free.

Proposition 0.4. :Let M be an A -module, then M is finitely generated $\Leftrightarrow$ there exists a finitely generated free A -module F and a surjective homomorphism $\varphi : F \rightarrow M$.

Proof:($\Rightarrow$): Since M is finitely generated, let $\{m_1, m_2, \dots, m_n\}$ be the generating set for M . Let $F = A^{(n)} = A \oplus A \oplus \dots \oplus A$ (n times), define

$$\varphi : F \rightarrow M$$

$$e_i \mapsto m_i,$$

where e_i stands for $(0, 0, \dots, 1, 0, \dots)$, then it's easy to check that φ is an epimorphism and F is a finitely generated free A -module;

($\Leftarrow$): suppose $\{s_1, s_2, \dots, s_n\} \subseteq F$ is the generating set for F . Since $\varphi : F \rightarrow M$ is an epimorphism, then for $\forall m \in M$, $\exists f \triangleq \sum_{i=1}^n a_i s_i$ such that $\varphi(f) = m$, where $a_i \in A$. Hence $m = \varphi(\sum_{i=1}^n a_i s_i) = \sum_{i=1}^n a_i \varphi(s_i)$ ($\forall m \in M$), which means $M \subseteq \langle \varphi(s_1), \varphi(s_2), \dots, \varphi(s_n) \rangle$, thus $M = \langle \varphi(s_1), \varphi(s_2), \dots, \varphi(s_n) \rangle$. M is finitely generated. $\square$

Proposition 0.5. :Let M be a finitely generated A -module, I is an ideal of A , $\varphi : M \rightarrow M$ is an A -module homomorphism such that $\varphi(M) \subseteq IM$. Then $\exists a_1, a_2, \dots, a_n \in I$ such that

$$\varphi^n + a_1 \varphi^{n-1} + \dots + a_{n-1} \varphi + a_n = 0 \quad (\in \text{Hom}_A(M, M) \triangleq \text{End}_A(M))$$

Proof: Suppose $\{x_1, x_2, \dots, x_n\}$ be a generating set for M , since $\varphi(x_i) \in IM$, hence $\varphi(x_i) \triangleq \sum_{j=1}^n a_{ij}x_j$ ($a_{ij} \in I$), i.e., $\sum_{j=1}^n (\delta_{ij}\varphi - a_{ij})(x_j) = 0$ ($i = 1, 2, \dots, n$). Written in matrix form, we have

$$\begin{pmatrix} \varphi - a_{11} & -a_{12} & \dots & -a_{1n} \\ -a_{21} & \varphi - a_{22} & \dots & -a_{2n} \\ \vdots & \vdots & & \vdots \\ -a_{n1} & -a_{n2} & \dots & \varphi - a_{nn} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = 0,$$

multiply the adjoint matrix of $(\delta_{ij}\varphi - a_{ij})$ on both sides, we get

$$\det(\delta_{ij}\varphi - a_{ij})x_i = 0$$

which means $\det(\delta_{ij}\varphi - a_{ij}) = 0$, expand the determinant we can get the equality of the form that $\varphi^n + a_1\varphi^{n-1} + \dots + a_{n-1}\varphi + a_n = 0$, where $a_1, a_2, \dots, a_n \in I$. $\square$

Corollary 0.1. :Let M be a finitely generated A -module, I is an ideal of A such that $IM = M$. Then $\exists x \in A$ such that $x - 1 \in I$ and $xM = \{0\}$.

Proof: Take $\varphi = Id : M \rightarrow M$, then $\varphi(M) = IM$, thus there exists $a_1, a_2, \dots, a_n \in I$ such that $Id + a_1 + a_2 + \dots + a_n = 0$, which means $(Id + a_1 + a_2 + \dots + a_n)m = 0$ ($\forall m \in M$). Let $x = 1 + a_1 + \dots + a_n$, hence $x - 1 \in I$ and $xM = \{0\}$. $\square$

By the corollary above, we have the following Nakayama Lemma

Nakayama Lemma

Theorem 0.2. :Let I be an ideal of A and $I \subseteq J(A)$, where $J(A)$ is the Jacobson radical of A . Let M be a finitely generated A -module, if $M = IM$, then $M = 0$.

Proof: Since $IM = M$, by the corollary above, $\exists x \in A$ such that $x - 1 \in I \subseteq J(A)$ and $xM = \{0\}$. Recall that $x \in J(A) \Leftrightarrow 1 - xy$ is a unit of A for $\forall y \in A$, hence $x = 1 - (1 - x) \cdot 1$ is a unit of A , which implies that $M = x^{-1}xM = \{0\}$. $\square$